Lab 1 Outline Final Draft

Team Diamond

1. Introduction

This section was written by: Priyaja Surendra

Societal Problem

1. Children spend a significant amount of time using digital devices for entertainment.
2. In 2021, children between the ages of 8 and 12 averaged approximately five hours and 33 minutes of entertainment screen use per day, compared with four hours and 44 minutes in 2019 (Rideout, 2022).
3. Excessive or unproductive technology use may negatively affect children's cognitive development, problem-solving skills, social interactions, and mental and emotional health (Lynn University, 2024).
4. Parents need an effective way to establish healthy screen-time limits while encouraging their children to use technology productively.

Needed Solution Characteristics

1. The solution should allow parents to establish and manage screen-time limits.
2. It should encourage children to complete educational activities.
3. Children should be able to earn additional screen time by completing educational tasks.
4. Parents should be able to customize questions and rewards according to their child's needs.
5. The solution should track each child's educational activity and progress.

Proposed Product

1. EduQuest is an educational screen-time management application designed for parents and children.
2. It allows children to earn additional screen time by completing educational questions and activities.
3. Parents can use EduQuest to manage screen-time limits, select or create educational questions, establish rewards, and review their child's progress.
4. The proposed prototype will demonstrate the Parent Controls Dashboard, educational questions, recorded screen-time rewards, and progress tracking. Device-wide screen-time enforcement is part of the real-world product vision, not the prototype.

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2. EduQuest Product Description

This section was written by: Tennya Boone

Product Overview

1. EduQuest is a parent-controlled screen-time management and learning platform for school-aged children and their parents.
2. Children answer short, age-appropriate educational questions to earn additional screen-time minutes.
3. Parents set question goals and reward amounts and review learning activity through the parent dashboard.
4. The real-world product envisions device-level screen-time management. The prototype focuses on a working learning-and-reward experience inside EduQuest.

Purpose and Benefits

1. Encourage children to use some recreational screen time to strengthen academic and problem-solving skills.
2. Connect educational practice with a reward children understand, rather than relying only on restrictions.
3. Help parents understand their child's progress through recorded answers, accuracy, and earned minutes.
4. Provide feedback after answers so children can learn from mistakes.

Prototype Scope

1. Use preconfigured demonstration accounts and a sample question bank instead of full account registration and a complete school curriculum.
2. Demonstrate parent settings, educational questions, answer validation, reward calculation, and basic learning-progress reporting.
3. Record earned minutes and demonstrate a timer and locking behavior only within a simulated entertainment environment in EduQuest. Earned minutes do not unlock the phone or other applications.
4. The current website demonstration uses in-memory data. Connecting the Python/FastAPI backend and PostgreSQL persistent storage is planned for CS 411.
5. OS-level app blocking and device-wide enforcement remain outside the prototype scope.

2.1. Key Product Features and Capabilities

This section was written by: Sincere Sanders

Significance

EduQuest transforms screen time from something that is simply restricted into a reward-based educational engagement tool. According to an article from Lynn University,

one potential disadvantage of technology at a young age is that it “potentially diminishes cognitive development and reduces problem solving skills” (Lynn University, 2024). EduQuest addresses this concern by requiring children to complete educational questions before earning additional screen time. This encourages children to spend more time learning while giving parents control over their child’s screen time limits. The intended value of EduQuest is connecting educational practice, parent-defined rewards, and progress reporting in one workflow.

Main Features & Capabilities

- **Parental Controls** - The Parent Controls Dashboard will allow the parent to set screen time limits for their child, determine how screen time can be earned, as well as the amount of extra time given for the completion of educational tasks. This allows parents to customize EduQuest to their child's needs.
- **Educational Questions** - EduQuest will provide a built-in database of educational questions covering a variety of subjects. The prototype uses a predetermined sample question bank; parent-authored questions remain a real-world product capability.
- **Screen Time Rewards** - This reward system records earned minutes after children meet the parent-defined correct-answer goal. In the prototype, those minutes apply only to the simulated entertainment environment within EduQuest. This feature creates a direct connection between completing educational work and receiving a reward.
- **Progress Tracking** - This progress tracking feature will record the child’s educational activity and performance, allowing the parent to monitor their child’s progress over time. This gives parents insight into how their child is performing and allows them to adjust settings in their dashboard based on their child's performance and needs.

2.2. Major Components (Hardware/Software)

This section was written by: Luke Morken

Prototype Hardware

1. **Client device:** A computer, tablet, or smartphone with a supported web browser displays the React Native Web interface. The prototype does not require a native iOS or Android application.
2. **Application server:** A server or development computer will run the planned Python/FastAPI API and communicate with the web interface over HTTP during development and HTTPS when deployed.
3. **Database server:** A standard PostgreSQL instance will provide persistent storage for the planned CS 411 prototype. A particular cloud provider or container deployment is not required by the prototype scope.
4. **Network connection:** An internet connection is needed for the hosted website; the integrated prototype will also need connectivity between the interface, API, and database.

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Selected Prototype Software

1. **Interface — JavaScript and React Native Web**

- Build on the existing browser-based parent dashboard and child interface rather than introducing another frontend framework (Gallagher and Meta Platforms).
- Display questions, answer feedback, earned minutes, settings, and learning progress.

2. **Backend API — Python and FastAPI**

- Planned for CS 411 integration; the current website demonstration is not connected to a backend.
- Validate answers, apply parent-defined reward rules, and return progress information to the interface ("FastAPI").
- Use parent-child relationships to determine which child's settings and learning records a parent may access. Preconfigured accounts replace full registration and login in the prototype.

3. **Persistent storage — PostgreSQL**

- Planned for CS 411 integration; the current demonstration stores data in memory.
- Store parent and child profiles, the sample question bank, screen-time settings, question attempts, learning records, and reward history ("About PostgreSQL").

4. **Build and package tools — Node.js, pnpm, and Vite**

- Node.js runs development tools and the existing JavaScript tests; pnpm manages packages; Vite runs the development server and builds the web interface ("About Node.js"; "Motivation"; "Getting Started").

5. **Version control and automated checks — Git, GitHub, and GitHub Actions**

- Track changes, review contributions, and run automated checks through the repository ("About Git"; "Understanding GitHub Actions").

6. **Testing**

- Use the existing Node.js test runner for frontend answer and reward logic.
- During CS 411, test the complete child-answer-to-parent-progress workflow and verify database updates, duplicate-reward prevention, and access boundaries. Integration testing should begin by weeks 9–10, with unit checks throughout development.

Component Interaction

1. The child submits an answer through the presentation layer.
2. The application layer validates the answer and calculates any earned minutes using the child's settings.
3. The data layer stores the attempt and resulting reward information in PostgreSQL once backend integration is complete.
4. The application returns updated results to the child interface and parent dashboard so both views reflect the same learning progress and reward balance.

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Future Product Components

1. Native mobile deployment, operating-system control integrations, and device-wide restrictions belong to the real-world product vision and require separate feasibility and permission work.
2. Firebase Cloud Messaging notifications, Docker deployment, and cloud scaling are possible future infrastructure choices, not required components of the selected prototype stack. The prototype displays progress directly in the parent dashboard.

3. Identification of Case Study

This section was written by: Justin Carmona

Primary Users

1. EduQuest is being developed for parents of children ages 8 to 12 who want to balance recreational screen time with academic reinforcement.
2. Parents use the dashboard to configure educational goals and rewards and review their child's answers, accuracy, and earned minutes.
3. Children use the question interface to complete educational practice, receive feedback, and see their reward progress.

Existing Tools and EduQuest's Focus

1. Apple Screen Time provides controls for device use, app access, and content (Apple).
2. Qustodio describes features including screen-time management, filtering, and activity reporting (Qustodio).
3. Bark offers parental monitoring and controls, including screen-time management and app or website blocking (Bark).
4. EduQuest's proposed focus is the question-to-reward workflow: children complete academic practice and parents see learning results and earned minutes. Its central demonstration connects a child's correct answers to earned minutes and a parent-visible progress record.

Why This Group

1. Ages 8 to 12 include elementary and upper-elementary learners who are strengthening reading and mathematics skills.
2. The Common Sense Census reports substantial recreational screen use among this age group (Rideout et al.).
3. Prototype evaluation will focus on the usability of the learning-and-reward workflow and the accuracy of recorded progress and earned minutes.

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Future / Expanded Users

Teachers or Tutors - Down the line, teachers could build or assign custom question sets tied to what's actually being covered in class that week, turning EduQuest into a home-reinforcement or test-prep tool rather than generic practice. A teacher-facing dashboard could show which students are consistently missing certain question types, flagging where extra help is needed.

Homeschool Families - Homeschool parents already act as both parent and educator, so they'd benefit from EduQuest's dual role: managing screen time and delivering self-paced, gradually adapting practice content. Progress tracking could double as informal record-keeping for their curriculum.

Expanded Age Ranges - The core mechanic of EduQuest is not inherently tied to ages 8-12. Content could scale down to early readers (K-2, sight words, basic counting) or up to middle schoolers (pre-algebra, reading comprehension), letting EduQuest grow with a family rather than being outgrown after a few years.

4. Glossary

This section was written by: Jessica Melton

Term	Definition
Academic Reinforcement	A practice that helps strengthen educational concepts a child is already learning.
API (Application Programming Interface)	A way for different parts of a software system to communicate and exchange information ("API").
Backend Server	The part of the system that handles data, requests, and other processes behind the scenes.
Built-in Question Database	A collection of educational questions provided by EduQuest.
Cross-Platform	Software that is designed to work on more than one operating system ("Introduction to React Native for Web").
Custom Questions	Custom educational questions created by parents.

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Parent Control Dashboard	The main area where parents manage screen-time limits, questions, rewards, and their child's progress.
Parental Controls	Settings that allow parents to manage how their child uses EduQuest, including screen-time limits and rewards.
PostgreSQL	The database system EduQuest will use to store account information and other application data (“About PostgreSQL”).
Progress Tracking	A feature that records a child’s educational activity and performance over time.
React Native Web	A library that makes React Native components and APIs available in web applications (Gallagher and Meta Platforms).
FastAPI	A Python framework selected for the planned backend API ("FastAPI").
Node.js	The JavaScript runtime used for project development tools and existing tests ("About Node.js").
pnpm	The package manager used to install and manage project dependencies ("Motivation").
Vite	The frontend development server and build tool ("Getting Started").
Recreational Screen Time	Time spent using digital devices mainly for entertainment, such as games or videos.
REST	A common way for applications to request, send, and receive information through an API (“REST”).
Screen-Time Limit	A parent-defined time allowance. The prototype demonstrates this only inside EduQuest; real-world device enforcement is future scope.

Screen-Time Management	The process of setting and managing how much recreational time a child can spend on a device.
Screen-Time Rewards	Earned minutes calculated from educational progress. The prototype records these minutes and uses them only in its simulated entertainment environment.
TLS (Transport Layer Security)	Technology used to protect information when it travels over the internet (Rescorla).

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